

PTVA's

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: B.Sc. Information Technology

Class: TY

Semester: V

Exam: ESE

Name of the Course: DevOps

Date of Exam: 06/10/2025

Marks: 60

Time: 2 hours

Instructions:

- (1) All questions are compulsory.
- (2) Make suitable assumptions wherever necessary and state the assumptions made.
- (3) Answers to the same question must be written together.
- (4) Numbers to the right indicate marks.
- (5) Draw neat labeled diagrams wherever necessary.
- (6) Use of non-programmable calculator is allowed.

Q 1 Attempt any TWO of the following:

12

- A. Define DevOps. Explain why culture plays a central role in DevOps adoption. Enumerate the three major cultural changes required for DevOps adoption.
- B. Break down the organizational changes required when banking on DevOps practices.
- C. Explain what is meant by "wicked" business problems in IT.
- D. Evaluate the significance of Lean waste removal strategies in sustaining DevOps transformation. Identify the eight elements of waste in Lean thinking.

Q 2 Attempt any TWO of the following:

12

- A. Use and explain API performance monitoring plays to solve issues in an e-commerce application.
- B. List and explain in brief the three components of the Agile Testing Trifecta.
- C. Judge the effectiveness on how the recommendations and action plan emphasize on due diligence in adopting an automated approach contributing to achieving continuous delivery best practices.
- D. Describe the major considerations in agile operations tooling.

Q 3 Attempt any TWO of the following:

12

- A. Examine how process time and lead time can be interpreted within a VSM and differentiate how each measure is utilized in the two general approaches - In-depth and Overview VSM.
- B. Formulate a transformation roadmap that mitigates the issues encountered when moving from choosing to adopting transformation plays.
- C. Describe the relationship between value propositions and revenue streams in the business model canvas.
- D. Specify the meaning of cycle time in context of DevOps delivery pipelines. Also explain the various cycle time embedded within the various other cycle times.

Q 4 Attempt any TWO of the following:

12

- A. Defend the use of Lean Startup methodologies in enterprise DevOps adoption.
- B. Clarify and explain why quality is considered everyone's responsibility in DevOps.

- C. Translate talent management hacks such as PIT apprenticeship programs and graduate intake into practical strategies for improving DevOps team performance.
- D. Compare Platform as a Service (PaaS) and Infrastructure as a Service (IaaS) for DevOps platform building.

Q 5 Attempt any TWO of the following:

12

- A. Discuss the purpose of developing a culture of continuous improvement.
- B. Apply the capabilities and goals of a DevOps CoC to accelerate enterprise-wide transformation. Show how core CoC roles can be applied to mentor and support DevOps adoption.
- C. Investigate how common DevOps transformation anti-patterns affect enterprise adoption and overall performance.
- D. Defend the approach of engaging the line of business for faster value delivery.